Report on DV14652

The manuscript studies radiative decays of heavy-strange axial-vector mesons using photon-DA light-cone sum rules, with particular emphasis on the suppression of $D_{s1}(2536) \to D_s\gamma$ through mixing. Testing this mechanism in LCSR could be a useful contribution. Since the qualitative mechanism is already discussed in Refs. [27,28], the significance of the present work depends on establishing its sum-rule origin and numerical reliability. I have two main concerns that should be addressed before publication in Physical Review D.

i) **Current mixing and separation of the physical states.** The state rotation in Eq. (6) does not by itself imply that the currents in Eq. (7) couple exclusively to the corresponding states, as assumed in Eqs. (8), (37), and (38). In general, both currents have residues $\langle 0|J^K_\mu|A_j\rangle = F_{Kj}\eta_\mu$, and the rotated residue matrix must be shown to be approximately diagonal. Both axial-vector states lie below the thresholds in Table II, and their small mass splitting provides little Borel suppression of the unwanted pole, especially in the extraction of the higher state.

There is also a concrete normalization issue: Eq. (3) of Ref. [29] prints the tensor current without the factor $1/(m_Q + m_s)$ introduced in Eq. (5) here, whereas Eq. (41) imports its mixing angles unchanged. A current-diagonalization angle depends on this relative normalization. Please establish the conversion and the relation to the state-mixing angle, and check the off-diagonal two-point correlator using the present OPE and current conventions throughout the working region. If the unwanted pole cannot be neglected, both poles should be retained. This is essential before identifying cancellation between two current contributions with physical-state interference.

ii) **Reliability of the suppressed amplitude.** The zero crossing in Fig. 1 makes the $D_{s1}(2536)$ prediction particularly sensitive to corrections that may be small in each basis amplitude. Please give the signed perturbative and individual twist contributions to $\widehat{\Pi}_A$, $\widehat{\Pi}_B$, and their physical combinations, together with quantitative continuum-suppression and OPE-convergence criteria for Table II. For the suppressed channel, compare the absolute size of the estimated omitted terms with the residual amplitude; convergence of the separate basis amplitudes is insufficient.

The stability of the suppression should also be tested against the assumed asymptotic leading-twist DA, the fixed twist-four parameters, and reasonable departures from $M_1^2 = M_2^2$. Please distinguish the robustness of the hierarchy of widths from that of the quoted sub-keV value, including an assessment of uncalculated radiative and higher-twist effects. A compact uncertainty budget would make the principal result substantially more convincing.

I also have the following specific comments. Equation, table, and reference numbers below refer to the manuscript unless stated otherwise.

1. **Continuum subtraction, p. 10 and Eqs. (32)–(35), (B2), (B6).** The same replacement $e^{-m_Q^2/M^2} \to e^{-m_Q^2/M^2} - e^{-s_0/M^2}$ is applied to every photon-DA term, including terms with different powers of M^2. This is not established merely by specifying quark–hadron duality. Please state the duality region in the two dispersion variables and derive the subtraction, including any boundary terms. Ref. [25] applies this replacement to the leading-twist term while leaving the higher-twist terms unsubtracted. Please explain the difference and quantify its effect on the cancellation-sensitive channel.

2. **Photon normalization and scales, Tables I and V.** The input in Ref. [30] is quoted at 2 GeV as $f_{\gamma,s}^{\perp} = -51.0(1.8)$ MeV. Please specify the evolution producing the quoted value at 1 GeV and the scales used for the quark masses and currents. More substantially, the finite-mass lattice result uses the subtraction $(1 - m_s\partial_{m_s})H_s$ [Ref. [30], Eqs. (17),(18)], which also changes finite mass terms. Please establish its compatibility with the photon-DA definition and the perturbative/soft separation in Eqs. (28)–(35), since strange-mass effects are explicitly retained.

3. **DA definitions and completeness, Eq. (20) and Appendix A.** Appendix A gives DA models but not the nonlocal matrix-element definitions promised below Eq. (33). These are needed to fix phases and normalization, particularly for the integrated vector DA $\psi^{(V)}$. Equation (A2) implies $\psi^{(V)}(1/2) = 0$, so Eq. (33) gives a vanishing axial-current twist-three term under Eq. (25), while the derivative term in Eq. (34) need not vanish. Please confirm this implementation and explain whether the unlisted axial two-particle and twist-three three-particle DAs vanish for the chosen projection or have been neglected.

4. **Uncertainty propagation, Sec. III and Fig. 3.** Please specify the sampling distributions and the two-point Borel windows and thresholds used to obtain f_i. Are the residues recomputed with the same sampled QCD inputs and mixing angle as the transition amplitudes, or sampled independently? Shared-input correlations should be retained. Please also clarify whether $f_i(\theta)$ is recomputed in Fig. 2, since the interpolating currents themselves change with θ.

5. **Momentum assignment, Eqs. (2), (8), and (15).** In Eq. (2), the Fourier momentum conjugate to the axial-current position is p, whereas Eq. (8) assigns p' to that channel. Please make the Fourier phases, current positions, and pole denominators consistent, including the momentum multiplying the tensor current. On p. 6, specify the spacelike virtualities and the hierarchy that justifies the light-cone expansion; the printed inequalities alone do not adequately define this regime.

6. **Auxiliary integral, Eq. (B4).** The first term integrates over α_s' although its integrand contains only α_s. Please check the primed arguments and associated denominators against the derivation and numerical implementation. Also define $\bar{u}_0$ explicitly or use the already defined a_0 consistently.

7. **Comparison with earlier LCSR results, Table IV.** Please check the pure axial-current transition kernel against Eq. (4.4) of Ref. [25] after translating conventions; the decay-constant cross-check at $\theta = 90^\circ$ does not test this kernel. A short comparison using common inputs would help distinguish the impact of mixing from changes in the photon normalization, residues, and continuum prescription behind the increase from 19–29 to approximately 46 keV.